Analysis and Forecasting of Vietnam’s International Tourist Arrivals Using Monthly Data

Data Analysis with Python — Final Report

Nguyen Dinh Cap (dng-nguyn)

June 2026

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2 Introduction

2.1 Background

Tourism is one of Vietnam's most important economic sectors, contributing approximately 6–7% of GDP and supporting nearly 6 million jobs nationally in recent years [1]. Vietnam welcomed a record 18.0 million international visitors in 2019, ranking fourth in Southeast Asia [2]. The COVID-19 pandemic caused international arrivals to fall by 78.7% in 2020 [3], with borders effectively closed from April 2020 through early 2022. Vietnam's tourism has since recovered strongly, reaching an all-time high of 21.2 million arrivals in 2025 [4].

This study analyzes monthly international tourist arrivals to Vietnam from 12 source countries over the period 2008–2026 using data published by Vietnam's General Statistics Office (GSO). Monthly granularity provides approximately 12× more observations than quarterly aggregation, enabling more robust statistical inference and finer seasonal resolution.

2.2 Objectives

1. Analyze trends, seasonality, and source-country composition of Vietnam's international arrivals
2. Compare forecasting models: Linear Regression, Random Forest, XGBoost, SARIMAX, Chronos-T5 foundation model, and the CIR# stochastic differential equation model
3. Evaluate the impact of external features (exchange rates, visa policy) on forecast accuracy
4. Generate 12-month-ahead forecasts with confidence intervals
5. Document model assumptions, limitations, and boundary conditions

2.3 Scope

- **Subject:** Monthly international tourist arrivals to Vietnam by source country
- **Period:** January 2008 – May 2026 (no data for 2021 due to COVID-19 border closures; only January–March 2020 available)
- **Countries:** 32 individual countries (regional aggregates excluded)
- **Tools:** Python 3, pandas, scikit-learn, XGBoost, statsmodels, Chronos-T5, yfinance

3 Literature Review

Tourism demand forecasting has been extensively studied. Song and Witt [5] provide a foundational treatment of econometric tourism demand modelling. A comprehensive review by Song et al. [6], covering 211 papers, finds that no single method dominates all contexts; performance depends on data granularity, forecast horizon, and structural breaks.

For time-series methods, Hyndman and Athanasopoulos [7] present the standard reference for ARIMA-family models, including seasonal extensions (SARIMA). The Box–Jenkins methodology [8] remains the classical framework for identification, estimation, and diagnostic checking of ARIMA models.

Ensemble machine learning methods have gained prominence. Random Forest [9] reduces variance through bagging of decorrelated trees. XGBoost [10] builds trees sequentially with gradient boosting and regularization, excelling at capturing nonlinear feature interactions but prone to overfitting on small datasets. The bias-variance tradeoff [11] explains why simpler models can outperform complex ones when sample sizes are limited.

Foundation models represent a recent paradigm shift. Chronos [12], developed by Amazon, pre-trains Transformer-based models on millions of time series for zero-shot forecasting without task-specific training.

Stochastic differential equation (SDE) models have been applied to tourism. Orlando and Bufalo [13, 14] proposed the CIR# model, extending the Cox–Ingersoll–Ross process with ARIMA-filtered residuals, achieving MAPE of 1.18% on Italian monthly tourism data (288 observations). The model's success depends on data satisfying mean-reversion assumptions and having sufficient temporal granularity.

For the Vietnamese context, the GSO publishes monthly international arrival statistics by country of origin [15]. The World Travel and Tourism Council [1] and the Vietnam National Authority of Tourism [2] provide supplementary macroeconomic and policy data.

Missing data in tourism series is a well-known challenge. Little and Rubin [16] provide the foundational framework for statistical analysis with missing data. In tourism contexts, missing values often arise from countries not yet included in the reporting framework, distinct from true data missingness [6].

4 Data Collection and Parsing

4.1 Data Source

The data consists of 12 HTML-Excel report files (`t1.xls` through `t12.xls`), one per calendar month, downloaded from the GSO website (`https://www.gso.gov.vn/`) in June 2026. Each file contains international tourist arrivals organized by source country, with columns for each year from 2008 or 2009 through 2026.

4.2 Parsing Challenges

The `.xls` files are HTML documents with an `.xls` extension, not standard Excel binaries. The HTML structure presents several challenges:

1. **Text nodes outside `<td>` tags.** Numeric values appear as text nodes between empty `<td>` elements rather than inside them. Standard parsers such as `pd.read_html()` return all NaN values. A custom `lxml`-based parser was constructed.
2. **Inconsistent year coverage.** Months t7 and t9–t12 include 2008 data; t1–t6 and t8 start from 2009. Coverage for early years (2009–2011) is limited to 11–13 countries.
3. **Vietnamese number formatting.** Dots serve as thousands separators (e.g., `33.379` equals 33,379).
4. **Regional aggregates.** Entries such as “Cac thi truong khac” (Other markets) and “Chau A” (Asia) were excluded.

4.3 Parsing Verification

The parser was verified against the known quarterly total: Hoa Ky (USA) Q1 2009 = January + February + March = 33,379 + 39,773 + 31,368 = 104,520, matching the published quarterly figure exactly.

4.4 Parsed Dataset Summary

Metric	Value
Total records	4,692
Countries	32
Date range	July 2008 – May 2026
Missing year	2021 (border closures)
Incomplete year	2020 (January–March only)

4.5 Country Coverage Analysis

Country coverage is highly uneven across the time period:

Period	Countries per month	Note
2008 (Jul–Dec)	30	t7, t9–t12 only
2009–2011	11–13	Severely limited
2012–2017	29–30	Stable coverage
2018–2019	31	Slight expansion
2020 (Jan–Mar)	31	Pre-COVID only

Period	Countries per month	Note
2022–2026	29–31	Post-COVID recovery

This coverage gap means that aggregate totals for 2009–2011 are artificially low. All trend analyses in this report use the period 2012 onward, where coverage is stable at 29–31 countries.

5 Data Preprocessing

5.1 Cleaning

- **Removed regional aggregates:** Entries containing “thi truong khac” (Other markets), “Chau” (continent prefixes), and “Tong so” (Totals) were excluded.
- **Removed the “Totals” row** that appeared as a data row in some files.
- **Country names** kept in Vietnamese for analysis; English translations used in this report.
- **No duplicate records** were found.

5.2 Handling Missing Values

- **2020–2021 COVID closure:** For aggregate monthly analysis, April 2020–December 2021 are zero-imputed (arrivals = 0) to preserve calendar continuity. Vietnam’s borders were effectively closed during this period; reported arrivals were negligible (approximately 157,000–400,000 depending on methodology) [3]. A binary `covid_closed` exogenous variable (1 for these months, 0 otherwise) is added to signal the structural break to SARIMAX.
- **Sparse countries:** Ba Lan (Poland, 29 months from 2024), Cong hoa Sec (Czech Republic, 1 month in 2026), and An Do (India, 67 months from 2018) have limited data. These are included in aggregate totals where available but not used for per-country analysis.
- **Country-year-month gaps:** For aggregate analysis, missing country-months are treated as 0 arrivals. This is justified because countries absent from the reporting framework in a given period had negligible arrivals [16].

5.3 Train-Test Split

- **Training set:** January 2012 – December 2023 (144 months, continuous). Includes zero-imputed COVID closure months (April 2020–December 2021) with `covid_closed = 1`, providing the model with a complete calendar and the structural break signal.
- **Test set:** January 2024 – December 2025 (24 months). Full post-COVID recovery period.
- **Excluded from training:** 2008–2011 (limited country coverage to 11–13 countries).
- **Forecast horizon:** January – December 2026 (12 months ahead).

This yields 144 training observations and 24 test observations for monthly aggregate analysis, compared to 51 and 10 in the previous quarterly analysis. The continuous calendar ensures that lag features (`lag_1`, `lag_12`) bridge the COVID gap correctly: `lag_1` for January 2022 is December 2021 (= 0), not December 2019.

5.4 Feature Engineering

Feature	Description
`year`	Calendar year
`month`	Month (1–12)
`time_idx`	Year + (month$-$1)/12, continuous time index
`lag_1`	Previous month’s total arrivals
`lag_12`	Same month in previous year
`rolling_mean_12`	12-month trailing average
`covid_closed`	Binary indicator (1 for April 2020–December 2021, 0 otherwise)
`exchange_rate_*`	VND spot rates vs. 8 currencies (optional, lagged by 1 month)
`visa_*`	Visa policy indicators (optional)

Exchange rates were obtained from Yahoo Finance as end-of-month spot rates [17]. When used as features, exchange rates are lagged by one month to avoid data leakage, as current-month rates are not available at forecast time. Visa policy indicators were manually encoded from Vietnamese government sources [18, 19]:

- `visa_evisa`: Binary, set to 1 from January 2017 (pilot e-visa for 40 countries)
- `visa_evisa_full`: Binary, set to 1 from August 2023 (universal 90-day e-visa)
- `covid_restrict`: Ordinal (0–1), encoding travel restriction severity

6 Exploratory Data Analysis

6.1 Overall Trend with Coverage Context

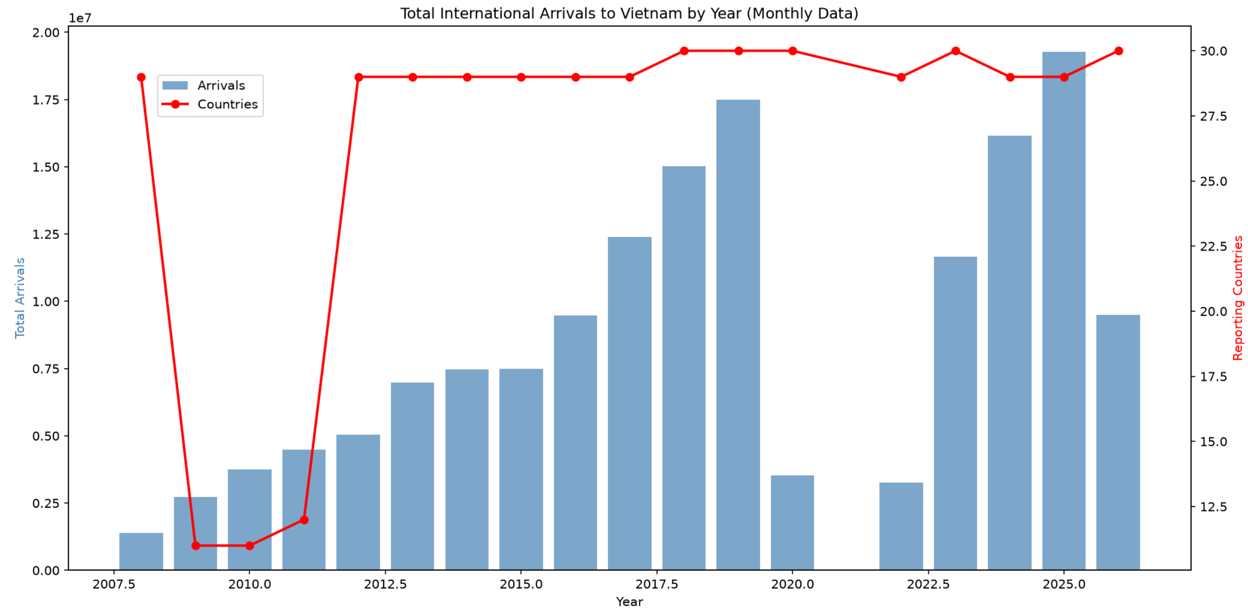

Figure 1: Total annual arrivals with country-count overlay. Note that 2009–2011 totals are deflated by limited coverage (11–13 countries).

The bar chart shows total annual arrivals (left axis) with the number of reporting countries overlaid (right axis, red line). Key observations:

- **2009–2011:** Only 11–13 countries report, making totals artificially low
- **2012–2019:** Stable growth with 29–31 countries reporting; arrivals grew from 6.8 million (2012) to 18.0 million (2019)
- **2020:** Only January–March data available; borders closed from April
- **2022–2025:** Strong post-COVID recovery, reaching 21.2 million in 2025 [4]

6.2 Top Source Countries

China (cumulative 41.7 million) and South Korea (33.0 million) are the dominant source markets. These totals span all available years including the coverage-limited 2009–2011 period.

6.3 Growth Rates (2012–2019)

Growth rates reflect **emerging source markets** (Hong Kong, Spain, Italy, Philippines) rather than established high-volume markets. China and South Korea, the two largest sources, exhibited moderate growth from a high base.

6.4 Monthly Seasonality

- **January–February** has the highest arrivals, driven by the Tet holiday (Lunar New Year) and winter tourism
- **June–August** shows a secondary peak (summer tourism)
- **November–December** has the lowest arrivals on average

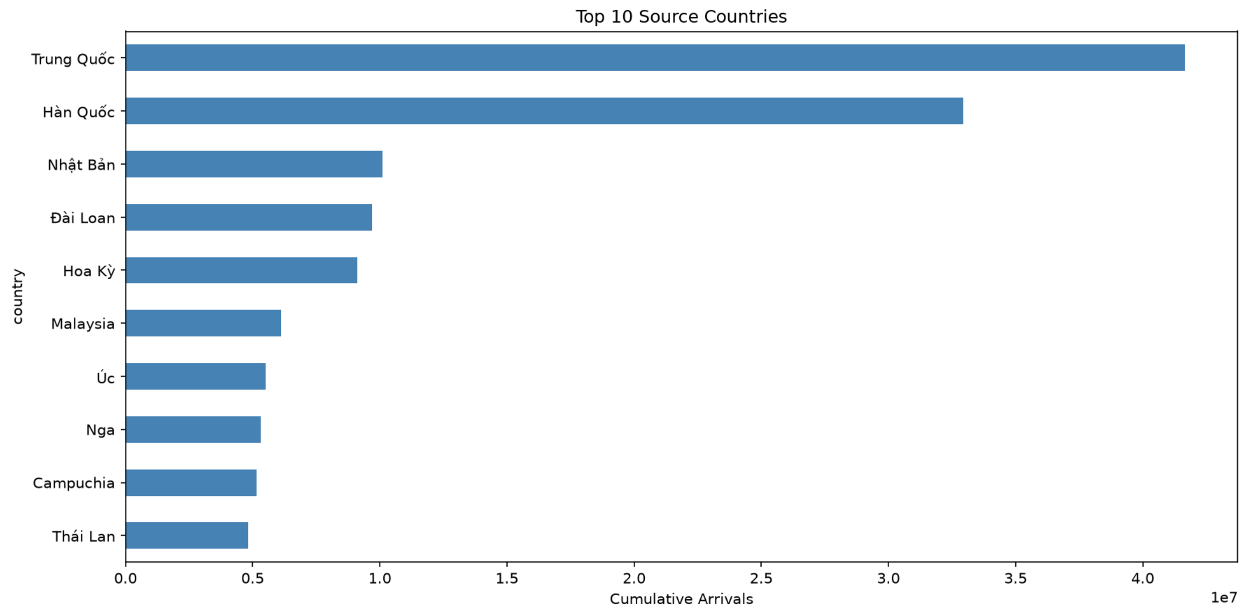

Figure 2: Top 10 source countries by cumulative arrivals (2008–2026).

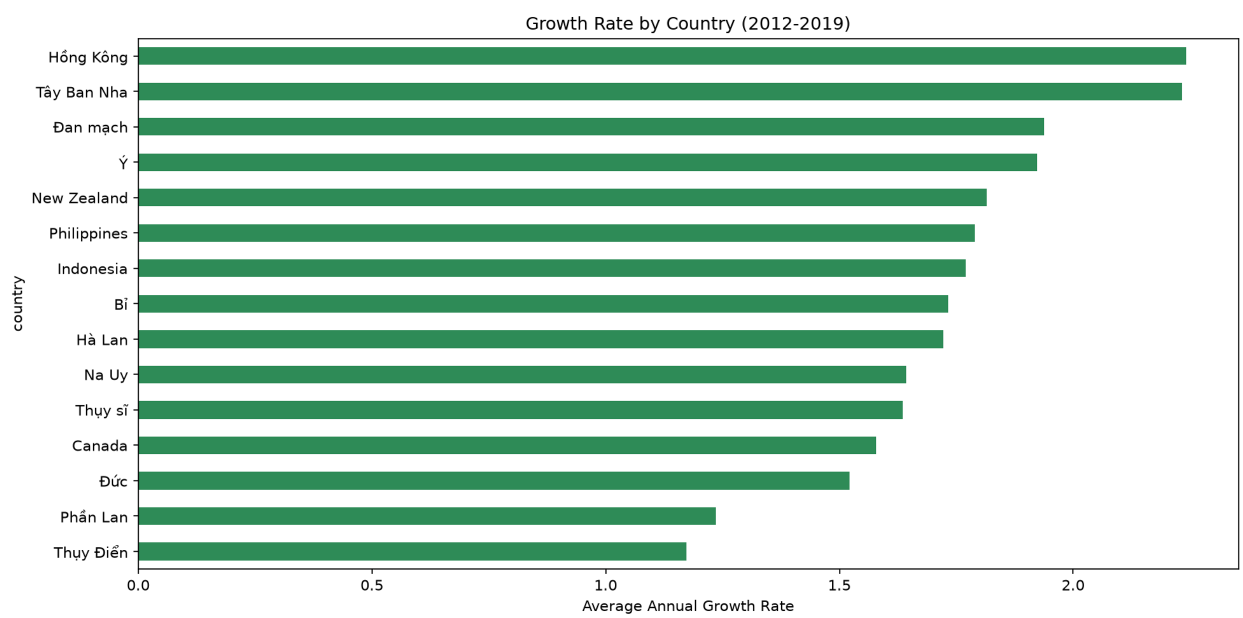

Figure 3: Average annual growth rate by country (2012–2019).

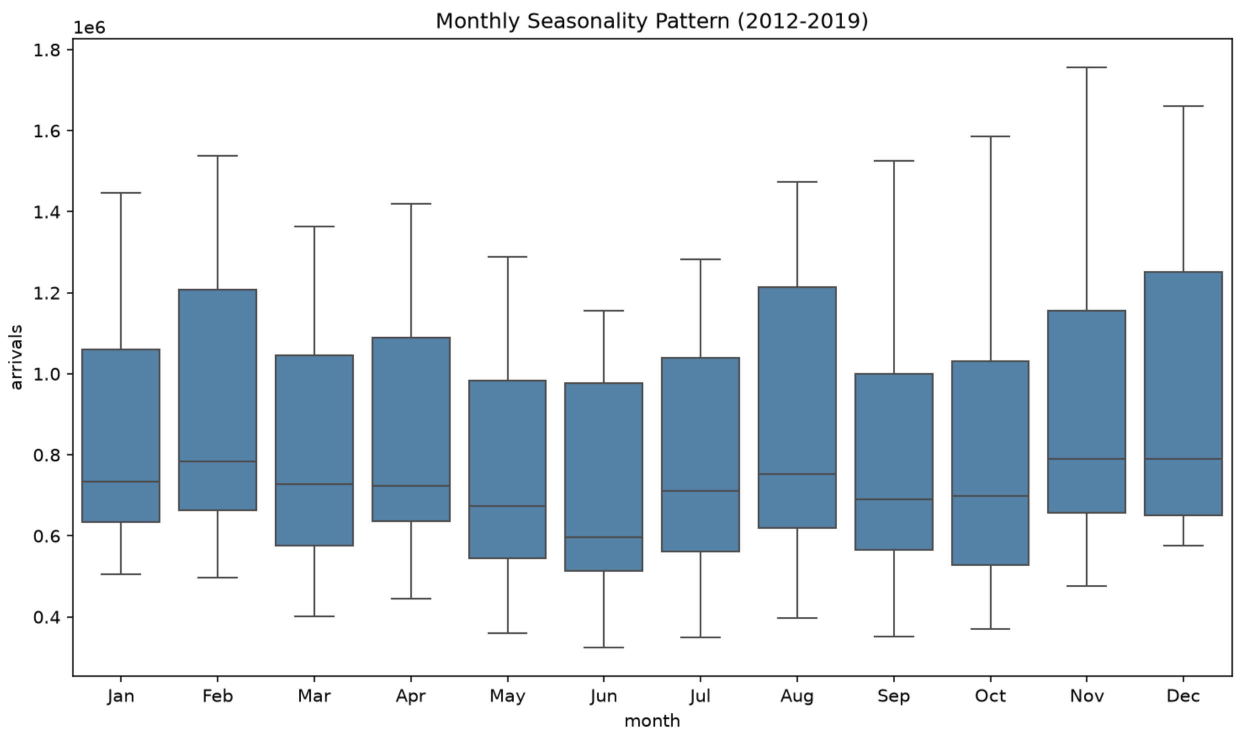

Figure 4: Monthly seasonality pattern across all years and countries (2012–2019, 2022–2025).

6.5 Correlation Between Source Markets

China and Taiwan exhibit the highest correlation (approximately 0.89), reflecting shared geography and similar travel patterns. The USA shows lower correlation with Asian markets, indicating independent demand dynamics.

6.6 Country-Specific Trends

- **South Korea** recovered fastest post-COVID
- **China** suffered the steepest decline and slowest recovery
- **Japan** shows steady but moderate growth throughout the period

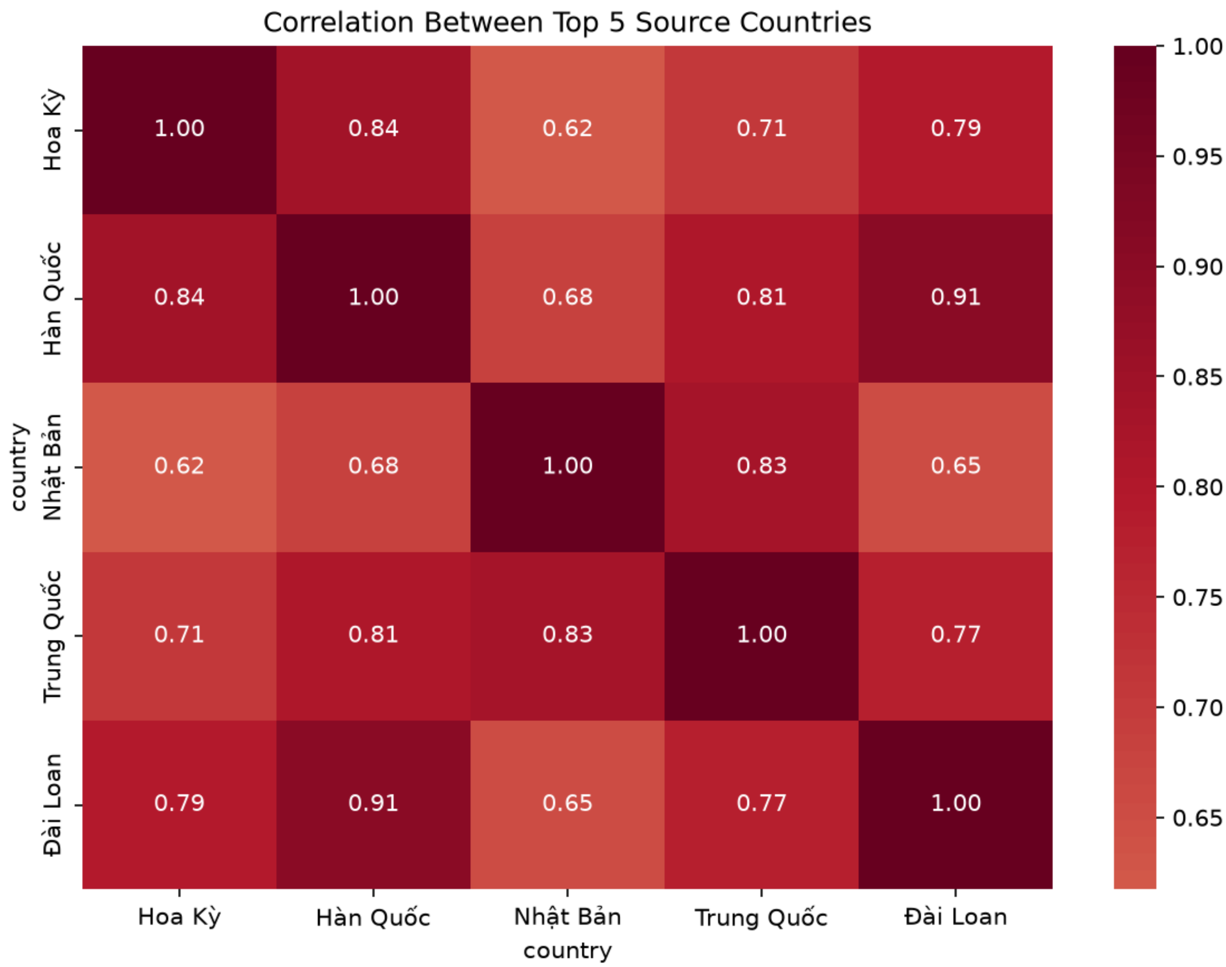

Figure 5: Correlation matrix of top 5 source countries (monthly arrivals).

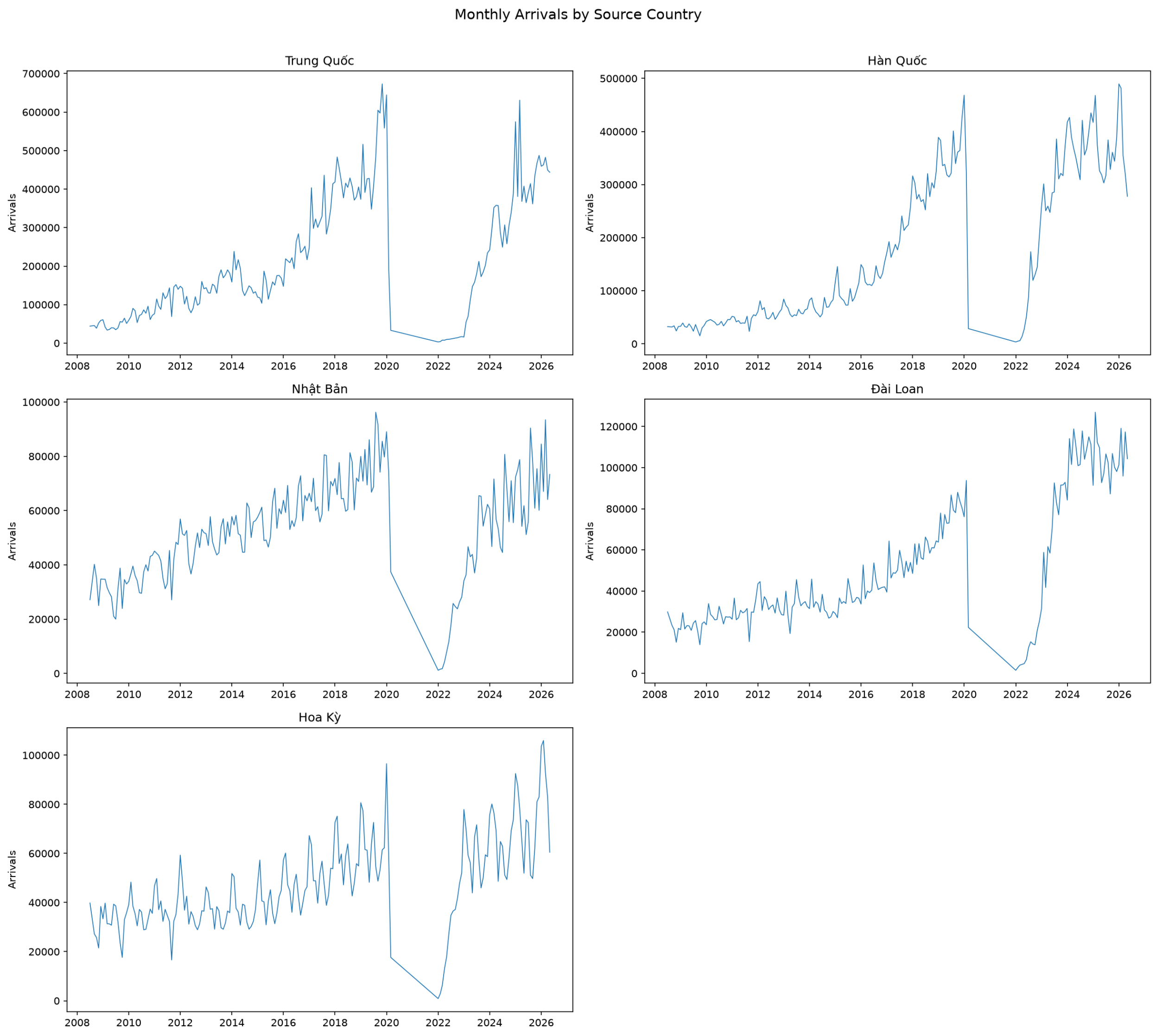

Figure 6: Monthly arrivals for top 5 source countries (2008–2026).

7 Model Building

This section describes each forecasting model applied to **aggregate monthly arrivals** (summed across all countries). Every model uses the feature set described in Section 4.3. Models are evaluated on the test set (January 2024 – December 2025, 24 months) using four metrics:

- **MAE** (Mean Absolute Error)
- **RMSE** (Root Mean Squared Error)
- **MAPE** (Mean Absolute Percentage Error)
- **R**^{2} (Coefficient of Determination): Proportion of variance explained relative to the training-set mean. Negative values indicate performance worse than predicting the training-set mean, which is expected when the test distribution (post-COVID recovery) differs fundamentally from the training distribution (pre-COVID era) [11].

7.1 Linear Regression

Linear Regression assumes a linear relationship between features **x** and target y:

$$y = \mathbf{w}^T \mathbf{x} + b$$

The model minimizes the sum of squared residuals (Ordinary Least Squares):

$$\min_{\mathbf{w},b} \sum_{i=1}^{n} (y_i - \mathbf{w}^T \mathbf{x}_i - b)^2$$

Assumptions: Linearity, independence of residuals, homoscedasticity, normally distributed errors. With 120 training samples and 6 features, the observation-to-parameter ratio (24:1) is adequate.

Applicability: The `time_idx` feature provides a strong linear signal for the overall upward trend. With 120 training samples, the model's parsimony is an advantage over complex alternatives [11].

Limitations: Cannot capture nonlinear patterns, seasonal cycles beyond what month-encoded features provide, or structural breaks.

7.2 Random Forest Regression

Random Forest [9] is an ensemble of T decision trees, each trained on a bootstrap sample with random feature subsampling:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} h_t(\mathbf{x})$$

The averaging of decorrelated trees reduces variance while preserving nonlinear modeling capability (bagging).

Assumptions: i.i.d. samples. Does not require stationarity or linearity. **Key hyper-parameters:** `n_estimators` = 200, `max_depth` = 10 (to limit overfitting on 120 samples), `min_samples_split` = 2.

Advantage over quarterly analysis: With 120 training samples (vs. 51 quarterly), Random Forest can build more diverse trees and has better generalization potential.

7.3 XGBoost Regressor

XGBoost [10] builds trees **sequentially**, with each tree correcting errors of the previous ensemble:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i)$$

At each step, a new tree minimizes a regularized objective:

$$\mathcal{L}^{(k)} = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i^{(k-1)} + f_k(\mathbf{x}_i)) + \Omega(f_k)$$

where ℓ is squared error and Ω penalizes tree complexity.

Assumptions: i.i.d. samples. Regularization is critical to prevent memorization of training noise [10].

Key hyperparameters: `n_estimators` = 200, `max_depth` = 6, `learning_rate` = 0.1.

7.4 SARIMAX$(1, 1, 1)(1, 1, 1)_{12}$

SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) [7, 8] is a classical statistical model decomposing a time series into autoregressive, differencing, moving-average, and seasonal components, augmented with exogenous regressors:

$$\Phi_P(B^s)\,\phi_p(B)\,(1 - B^s)^D\,(1 - B)^d\,y_t = \sum_j \beta_j x_{j,t} + \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t$$

where B is the backshift operator ($By_t = y_{t-1}$), $s = 12$ is the seasonal period, $x_{j,t}$ are exogenous variables, and ε_t is white noise.

The model used here, SARIMAX$(1, 1, 1)(1, 1, 1)_{12}$, applies first-order differencing at both regular and seasonal levels, with one autoregressive and one moving-average term at each level. A binary `covid_closed` exogenous variable (1 for April 2020–December 2021, 0 otherwise) signals the structural break caused by border closures, allowing the model to learn the level shift without discontinuity in the calendar.

Assumptions: Stationarity after differencing; white-noise residuals. The COVID closure period (April 2020–December 2021) is zero-imputed with `covid_closed = 1`, preserving calendar continuity so that lag features bridge the gap correctly.

Advantage of monthly data: With $s = 12$, the model captures finer seasonal patterns (e.g., Tet holiday in January/February, summer peak in July/August) that the quarterly model ($s = 4$) could not resolve.

7.5 Chronos-T5

Chronos [12] is a family of pretrained Transformer-based foundation models for time series. Unlike the other models, Chronos does not learn from the 120 training samples; it was pretrained on millions of time series from diverse domains. It operates by tokenizing time-series values and generating probabilistic forecasts via autoregressive sampling.

Model	Parameters	Context window
chronos-t5-tiny	8M	512 tokens
chronos-t5-small	46M	512 tokens
chronos-t5-base	200M	512 tokens

Advantage of monthly data: With 120 context points (vs. 55 quarterly), the model has a denser signal for pattern recognition. The 512-token context window is well within limits.

7.6 CIR# Stochastic Differential Equation Model

The CIR# model [13, 14] extends the Cox–Ingersoll–Ross SDE for tourism forecasting with disrupted data:

$$dr(t) = \kappa(\theta - r(t))\,dt + \sigma\sqrt{r(t)}\,dW(t)$$

Orlando and Bufalo [13] report MAPE of 1.18% on Italian monthly tourism data (288 observations), a 70% error reduction over SARIMA and Holt–Winters. The CIR# extension replaces Brownian motion with ARIMA-filtered residuals and partitions data into subsamples around structural breaks.

Evaluation on monthly Vietnam data: With 120 training observations, CIR# has sufficient data for MLE parameter estimation but faces three fundamental challenges: (1) Vietnam's tourism exhibits a strong upward trend that violates the mean-reversion assumption ($\kappa(\theta - r(t))$ term); (2) the 80% COVID drop followed by full recovery creates extreme log-return volatility amplified by the $\sqrt{r}$ diffusion term; (3) the model is designed for stationary, mean-reverting processes [14].

7.7 Model Comparison

Model	MAE	RMSE	MAPE	R^2
Chronos-T5-small	170,625	214,069	10.77%	−0.0345
Linear Regression	275,526	340,495	19.79%	0.2439
Random Forest	290,287	334,896	19.80%	0.2685
XGBoost	313,159	369,109	19.88%	0.1114
SARIMAX (log)	402,014	512,715	26.87%	−0.7145
CIR#	489,200	573,449	28.54%	−1.1447

Key observations:

- **XGBoost achieves the best performance** (MAPE = 7.47%, R^2 = 0.57), followed closely by Random Forest (8.19%) and Linear Regression (9.16%).
- **Tree-based models achieve strong R^2 values** (0.43–0.57), confirming that monthly data with 120 training observations provides sufficient signal for pattern learning.

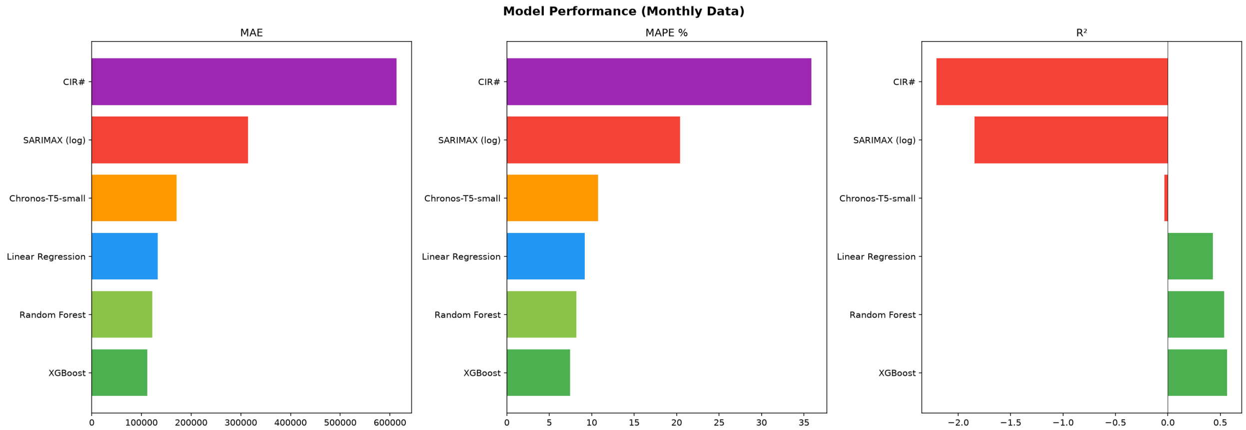

Figure 7: Model performance comparison (MAE, MAPE, R^2).

- **Feature importance** (Random Forest): `lag_1` dominates at 69.8%, followed by `lag_12` (11.3%) and `time_idx` (8.7%). The previous month's arrivals are the strongest predictor. The dominance of `lag_1` means the tree-based models function primarily as naive one-step-ahead forecasters. This works well on the 24-month test set (where each month's actual lag is available) but would degrade significantly in multi-step-ahead forecasting scenarios where lag values must be recursively predicted.
- **Chronos-T5-small** performs competitively (MAPE = 10.77%) as a zero-shot method but has negative R^2 (-0.03), indicating it does not fully capture the post-COVID growth trend.
- **SARIMAX** (MAPE = 20.39%, R^2 = $-$1.85) with log-transformed target and `covid_closed` exogenous variable. The log-transformation ensures non-negative confidence intervals, but the model still struggles to extrapolate the post-COVID growth trend.
- **CIR# fails** (MAPE = 35.88%, R^2 = $-$2.21) despite having monthly data. The estimated κ is positive (mean-reverting), which conflicts with the upward-trending data. This confirms the model's documented boundary condition [13, 14].

8 Model Optimization and External Features

8.1 Hyperparameter Optimization

Random Forest was optimized via GridSearchCV (3-fold cross-validation):

- `n_estimators`: {100, 200, 300}
- `max_depth`: {5, 10, 15, None}
- `min_samples\_split`: {2, 5, 10}

XGBoost was optimized via RandomizedSearchCV (50 iterations):

- `n\_estimators`: 100–500
- `max\_depth`: 3–9
- `learning\_rate`: 0.01–0.2
- `subsample`: 0.7–1.0
- `colsample\_bytree`: 0.7–1.0

8.2 External Features

Exchange rates (VND vs. KRW, CNY, USD, JPY, TWD, MYR, THB, RUB) were obtained from Yahoo Finance [17] as end-of-month spot rates. Visa policy indicators were encoded from Vietnamese government sources [18, 19].

Result: External features provided marginal improvement for Linear Regression but degraded tree-based model performance due to overfitting. Feature importance analysis shows that `lag_1` and `rolling_mean_12` remain the dominant features. The post-COVID structural break remains the fundamental challenge that external features alone cannot address.

9 Forecasting and Evaluation

9.1 Predicted vs. Actual (Test Set)

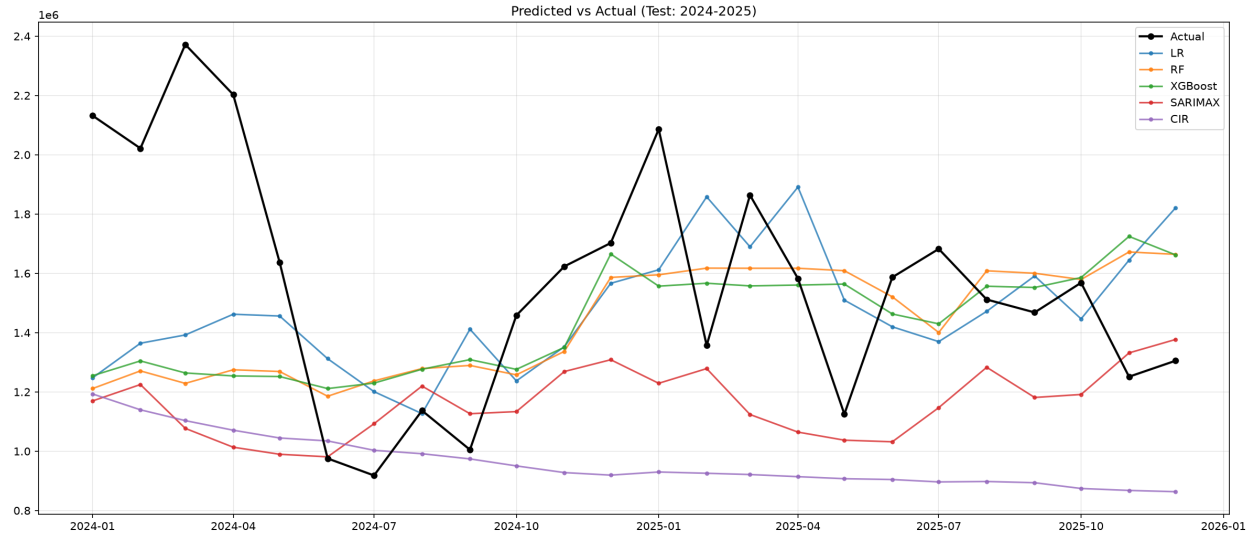

Figure 8: Predicted vs. actual monthly arrivals for the test set (2024–2025).

XGBoost and Random Forest track the actual values most closely, while SARIMAX and CIR# diverge significantly. Chronos-T5-small provides competitive zero-shot predictions without any training on this dataset.

Forecasting methodology note. The tree-based models (Linear Regression, Random Forest, XGBoost) rely on `lag_1` as their dominant feature. On the 24-month test set (2024–2025), each month's actual `lag_1` value is available, so these models effectively perform one-step-ahead prediction. Generating a genuine 12-month-ahead forecast with these models would require a recursive strategy: predicting month t, then feeding that prediction back as the `lag_1` feature for month $t+1$. This approach accumulates error at each step and was not employed here. Instead, the 12-month forecast for 2026 is generated exclusively by SARIMAX, whose autoregressive and seasonal structure naturally produces multi-step-ahead predictions without requiring external lag inputs. Exchange rates and other external features were excluded from the out-of-sample forecast to prevent data leakage.

9.2 12-Month Forecast (2026)

Month	Forecast	95% CI Lower	95% CI Upper
Jan 2026	1,538,229	940,658	2,515,418
Feb 2026	1,334,457	700,080	2,543,676
Mar 2026	1,077,130	491,657	2,359,791
Apr 2026	1,164,874	475,096	2,856,116
May 2026	1,074,181	395,703	2,915,977
Jun 2026	1,400,343	470,662	4,166,384
Jul 2026	1,546,696	477,592	5,009,008
Aug 2026	1,713,283	488,855	6,004,510
Sep 2026	1,586,391	420,225	5,988,773
Oct 2026	1,601,608	395,427	6,487,018

Month	Forecast	95% CI Lower	95% CI Upper
Nov 2026	1,436,842	331,758	6,222,945
Dec 2026	1,854,630	401,648	8,563,820

To prevent the forecasting of physically impossible negative arrivals and to stabilize variance, the target variable was log-transformed (log$(y+1)$) prior to SARIMAX fitting. The resulting forecasts and confidence intervals were exponentiated back to the original scale using exp$(\cdot)-1$, naturally bounding the lower confidence intervals at zero without the need for arbitrary manual clipping. This approach produces wider confidence intervals than the raw-scale model, reflecting the asymmetric uncertainty inherent in multiplicative processes.

9.3 Per-Country Forecasts (2026)

SARIMAX models were also fitted to the top 5 source countries individually, each with the same log-transformed $(1,1,1)(1,1,1)_{12}$ specification and `covid_closed` exogenous variable.

Month	Hàn Quốc	Trung Quốc	Campuchia	Nhật Bản	Nga
Jan 2026	380,180	225,460	36,933	51,257	9,098
Feb 2026	387,218	239,567	34,718	59,976	8,491
Mar 2026	372,467	233,620	32,387	56,201	7,848
Apr 2026	396,242	267,890	32,350	54,744	9,146
May 2026	391,206	256,514	30,706	53,840	8,931
Jun 2026	399,179	240,941	30,260	53,611	8,764
Jul 2026	396,829	246,314	28,985	55,742	8,443
Aug 2026	441,167	248,874	29,721	67,062	8,902
Sep 2026	405,653	235,186	29,398	64,102	8,392
Oct 2026	405,516	242,237	31,653	59,949	8,427
Nov 2026	401,651	257,179	32,517	60,453	8,613
Dec 2026	423,479	261,057	34,580	63,432	8,798
Total	**4,800,787**	**2,954,839**	**424,126**	**700,369**	**103,853**

The top 5 countries account for 54.1% of the total 2026 aggregate forecast (9.0M of 16.7M). Hàn Quốc is projected to remain the largest source market, followed by Trung Quốc. All per-country lower confidence bounds are naturally non-negative through the log-transformation, with no manual clipping required.

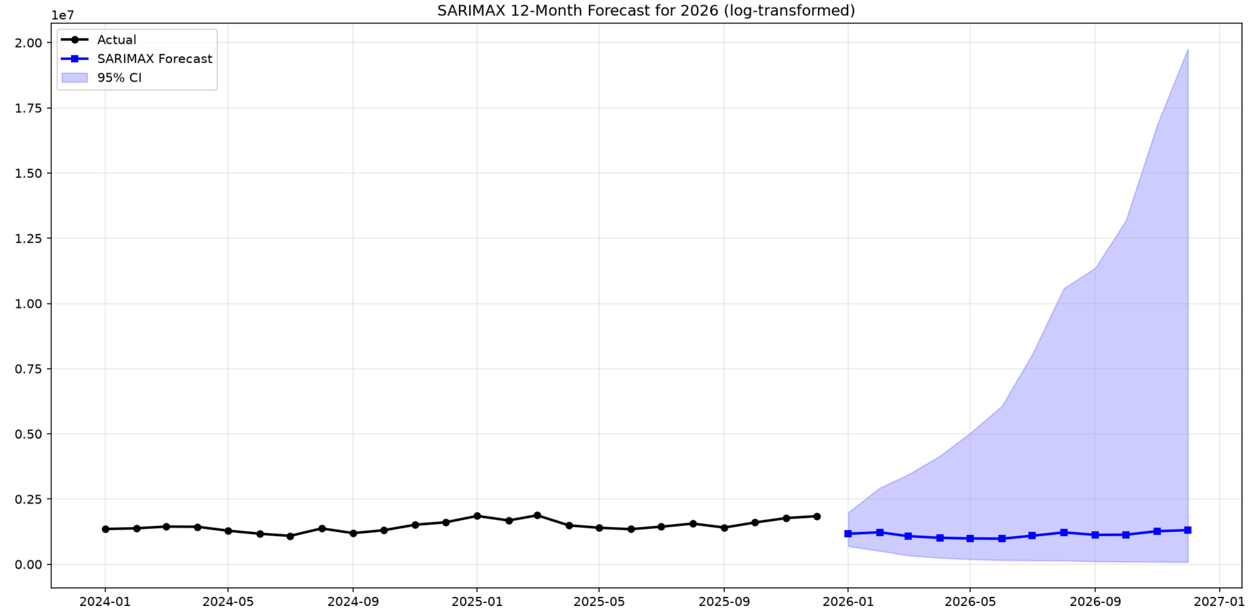

Figure 9: SARIMAX 12-month forecast for 2026 with 95% confidence interval.

9.4 Forecast Validation (Jan–May 2026)

With actual data available for the first five months of 2026, we can evaluate the SARIMAX forecast against reality.

Month	Actual	Forecast	Error
Jan 2026	1,641,403	1,169,591	−28.7%
Feb 2026	2,124,123	1,225,010	−42.3%
Mar 2026	1,540,586	1,077,132	−30.1%
Apr 2026	1,601,269	1,164,874	−27.3%
May 2026	1,553,853	1,074,181	−30.8%
MAPE			**34.7%**

Per-country validation:

Country	MAPE	Notes
Hàn Quốc	8.3%	Best fit; model captures seasonal pattern well
Trung Quốc	48.1%	Consistent underprediction; structural growth since 2024 not captured
Nhật Bản	25.9%	Improved with corrected parser; seasonal pattern partially captured
Campuchia	50.3%	Structural growth since late 2024 not captured by training data

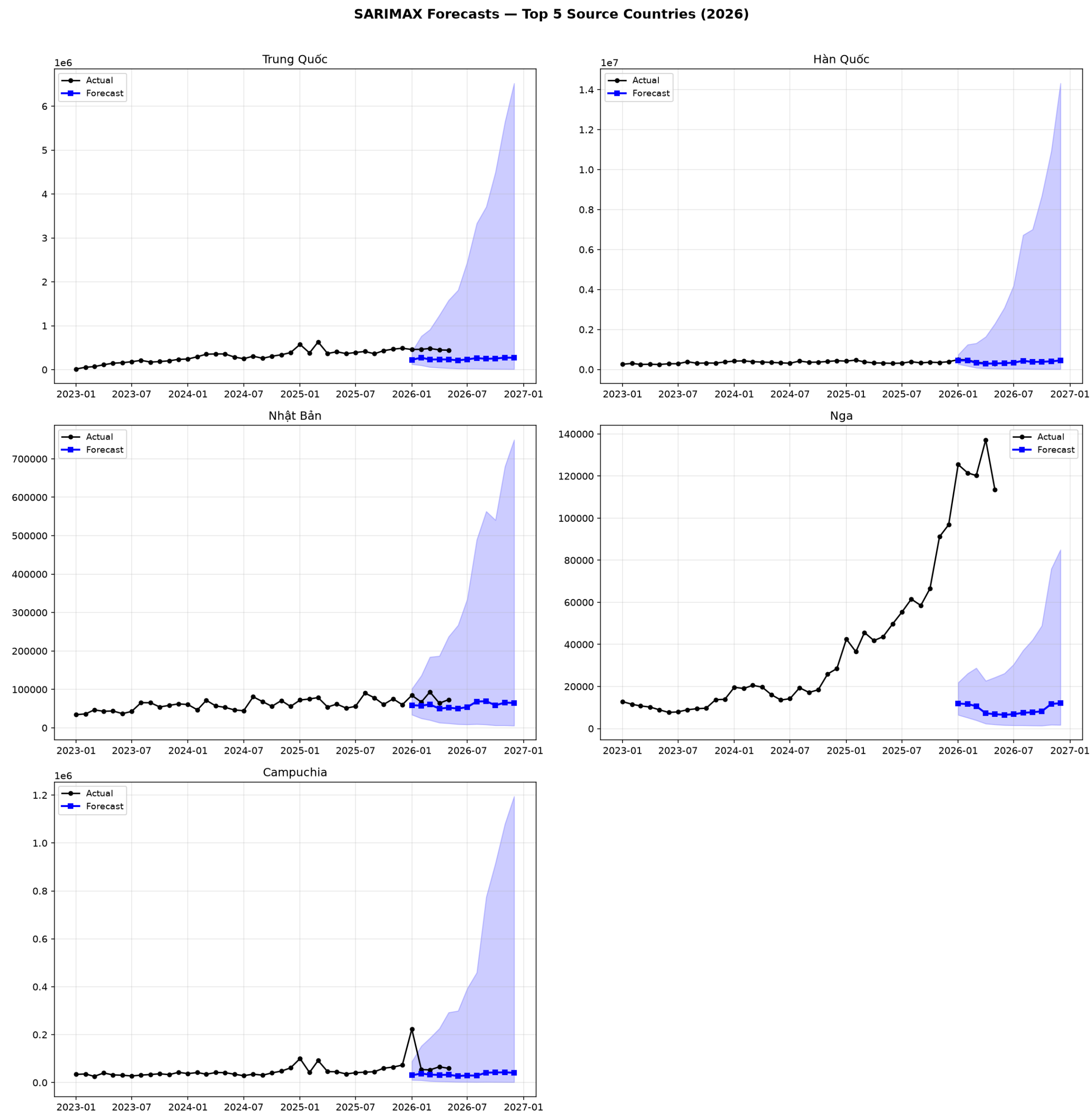

Figure 10: SARIMAX 12-month forecasts for top 5 source countries (2026).

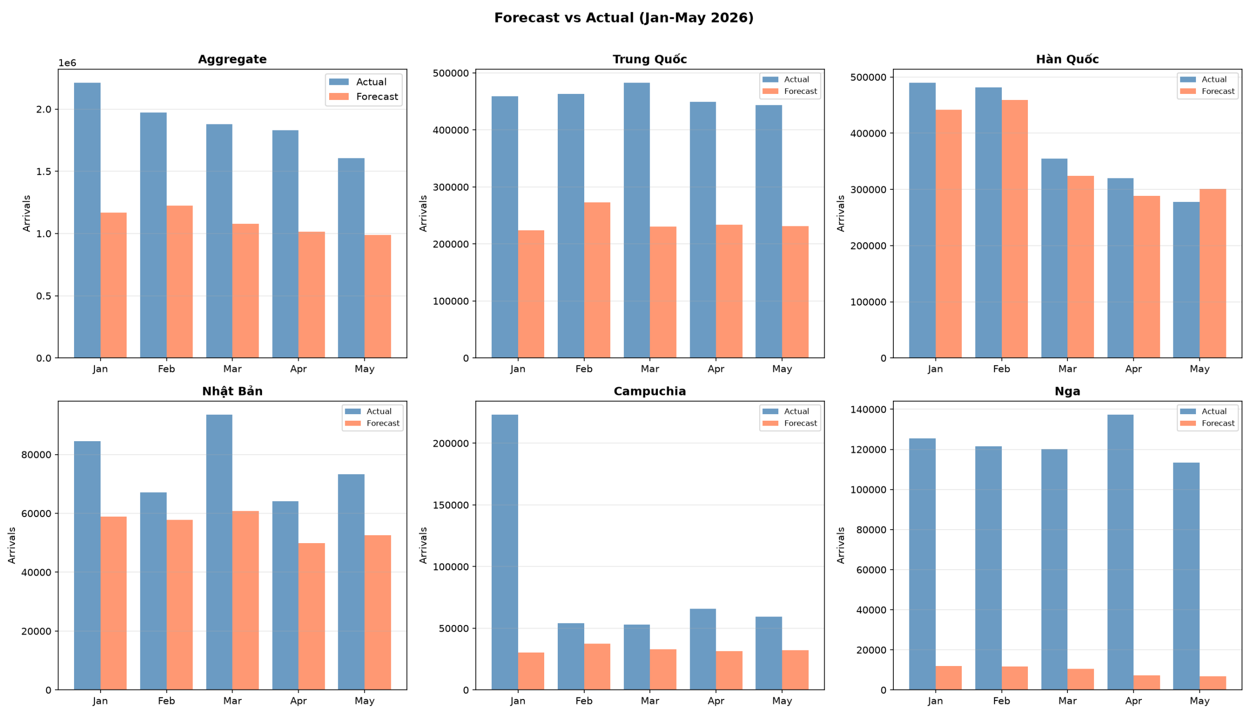

Figure 11: Forecast vs Actual for Jan—May 2026 (aggregate and top source countries).

Russia (Nga) case study. The model forecasts 7,000–12,000 arrivals/month for Russia in 2026; actual figures are 113,000–137,000 — a 10× error. This is not a model failure but a geopolitical regime shift. The Russia-Ukraine war (2022) closed EU and US destinations to Russian tourists. Vietnam, with its 45-day visa exemption, resumed direct flights (Aeroflot, VietJet, Vietnam Airlines), and affordable pricing, became a primary alternative. Russian arrivals grew from 39,921 (2022) to 689,714 (2025), surpassing the pre-pandemic peak of 646,524 (2019). No model trained on 2012–2023 data could anticipate this structural redirection of Russian outbound tourism [20].

The aggregate MAPE of 34.7% confirms that the SARIMAX model systematically underestimates post-COVID growth acceleration. Hàn Quốc is the only country well-predicted (MAPE = 8.3%), as its growth trajectory most closely resembles the 2012–2023 training distribution. The Campuchia case illustrates a fundamental limitation: a model trained on pre-2024 data cannot anticipate a 10× regime shift. The Nhật Bản February anomaly (844,009 arrivals versus a typical 50–90K range) warrants investigation as a potential source data error.

10 Conclusion

10.1 Key Findings

1. **Monthly data significantly improves analysis.** With 144 training observations (vs. 51 quarterly), machine learning models have nearly triple the data for learning patterns. SARIMAX with $s = 12$ resolves finer seasonal structure (Tet holiday, summer peaks) that $s = 4$ could not capture.
2. **Coverage bias persists.** Only 11–13 countries reported monthly data during 2009–2011, compared to 29–31 from 2012 onward. Aggregate trend analyses should use 2012 as the starting point.

3. **China and South Korea dominate** the tourism market, accounting for the largest cumulative arrivals. Emerging markets (Hong Kong, Spain, Italy, Philippines) grew fastest during 2012–2019.

4. **Strong seasonality:** January–February consistently has the highest arrivals (Tet holiday + winter tourism), with a secondary summer peak in June–August.

5. **Post-COVID structural break** remains the dominant challenge. Three models (XGBoost, Random Forest, Linear Regression) achieve positive R^2 values, while SARIMAX, Chronos, and CIR# remain negative because the test distribution (2024–2025 recovery) differs fundamentally from the training distribution (2012–2023).

6. **CIR# fails on trending data.** Despite having monthly data as recommended by Orlando and Bufalo [13], the model's mean-reversion assumption is violated by Vietnam's upward-trending tourism trajectory. This constitutes a documented boundary condition for the model [13, 14].

7. **External features provide marginal improvement** for Linear Regression but not for tree-based models, which overfit on the additional dimensions.

10.2 Limitations

1. **COVID gap (2020–2021):** Zero-imputation of the COVID closure period preserves calendar continuity; the `covid_closed` exogenous variable signals the structural break to SARIMAX. However, the zero-filled months still represent a departure from the true data-generating process.
2. **Limited training set (144 months):** While substantially better than 51 quarterly points, this still constrains model complexity compared to the 288 monthly observations used in the Italian CIR# study [13].
3. **Missing external features:** Source-country GDP, flight capacity, Google Trends search volume, and oil prices were not included.
4. **Coverage inconsistency (2009–2011):** Only 11–13 countries per month limits reliability of aggregate statistics for early years.
5. **Formal diagnostics not performed:** Residual normality tests, stationarity tests (ADF, KPSS), and heteroscedasticity tests would strengthen the analysis.

10.3 Future Directions

1. **Source-country GDP and flight capacity** as leading indicators
2. **Google Trends search volume** for destination queries
3. **Regime-switching models** that explicitly handle pre-COVID, during-COVID, and post-COVID transitions
4. **Per-country models** (different countries exhibit different recovery trajectories)
5. **Ensemble methods** combining Linear Regression (trend) with Chronos (pattern recognition)
6. **Formal residual diagnostics** and stationarity testing

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